



AI-Driven Archaeological Site Mapping

By

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Introduction to AI-Driven Archaeological Site Mapping

- This project presents an **AI-powered system** for archaeological site analysis using satellite and terrain data.
 - The platform integrates **deep learning models** to: Detect archaeological artifacts, Segment terrain features, Predict soil erosion risk
 - Designed to support **archaeologists, researchers, and planners** in early-stage site exploration.
 - Focuses on **automation, accuracy, and visual decision support**.
-

Technology Stack

Framework

Streamlit

Deep Learning

- PyTorch
- Segmentation Models
PyTorch (SMP)

Computer Vision

- OpenCV
- YOLOv5

Machine Learning

- Scikit-Learn
(Random Forest)

Data Processing

- NumPy
- Pandas

Visualization

- Matplotlib
 - PIL
-

Challenges

1. Manual archaeological surveys

Traditional surveys are time-consuming, costly, and difficult to scale across large or remote areas.

2. Terrain degradation and erosion

Erosion and landslides can damage or destroy archaeological sites before they are discovered.

3. Limited use of AI

Existing satellite and geospatial data are underused due to limited AI-based analysis.

Core Features

Automated Artifact Detection

AI models automatically identify archaeological artifacts from images.

Predictive erosion analysis

Machine learning predicts erosion risk using environmental data.

Terrain and Land-feature segmentation

Deep learning separates terrain features from satellite imagery.

Unified visualization dashboard

A single dashboard displays all model outputs clearly.

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Semantic Segmentation

- U-Net trained on satellite and DEM data
- Outputs pixel-level terrain classification

Object Detection

- YOLOv5 trained on annotated artifact images
- Detects and classifies archaeological objects

System Architecture

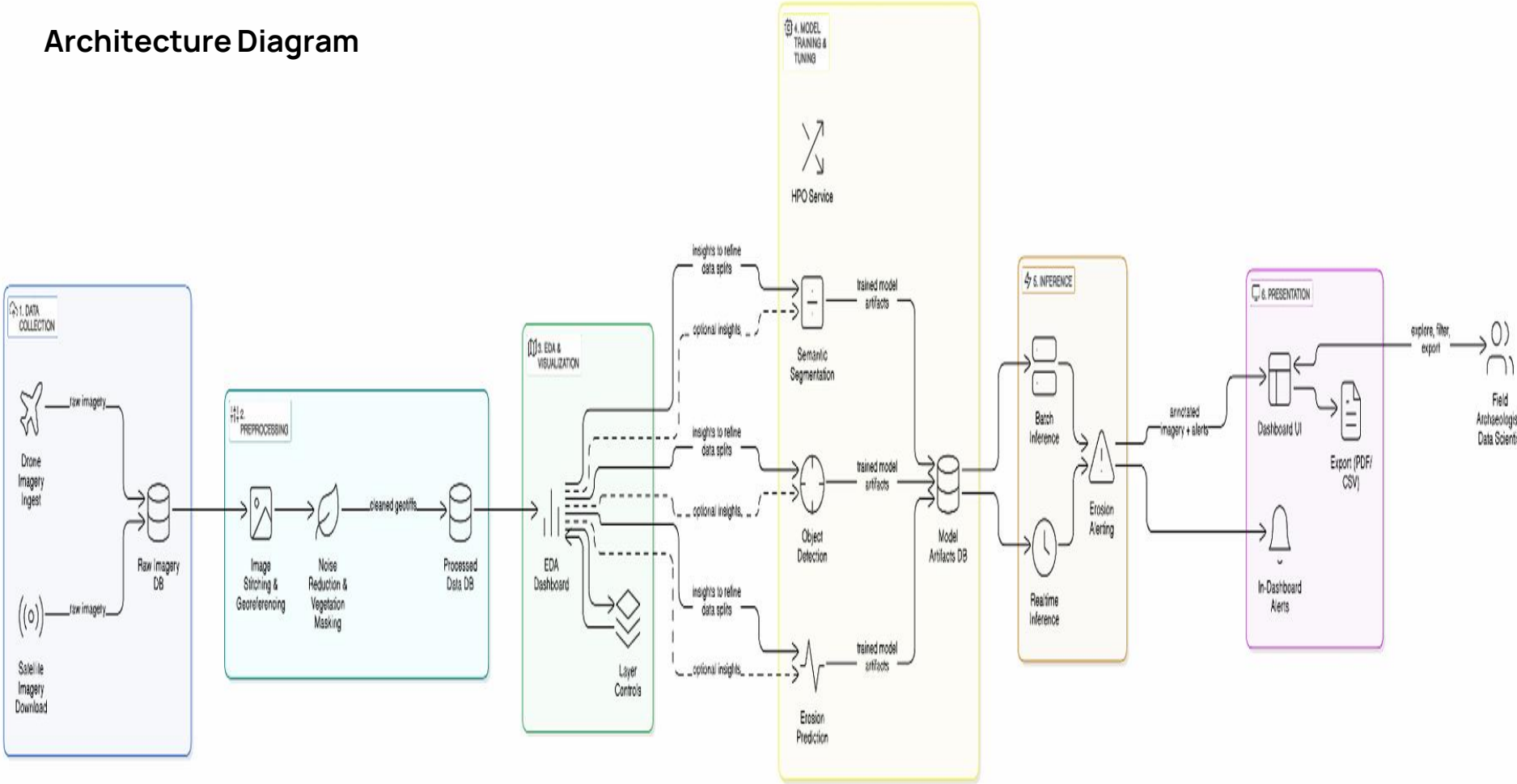
Erosion Prediction

- Regression-based ML model
- Uses NDVI, slope, DEM, and climate features

Deployment

- Streamlit-based interactive dashboard

Architecture Diagram



YOLOv5 Artifact Detection App

Upload an image to detect historical artifacts (coin, jewelry, pottery, sculpture, seal, tablet, weapon).

Choose an image...



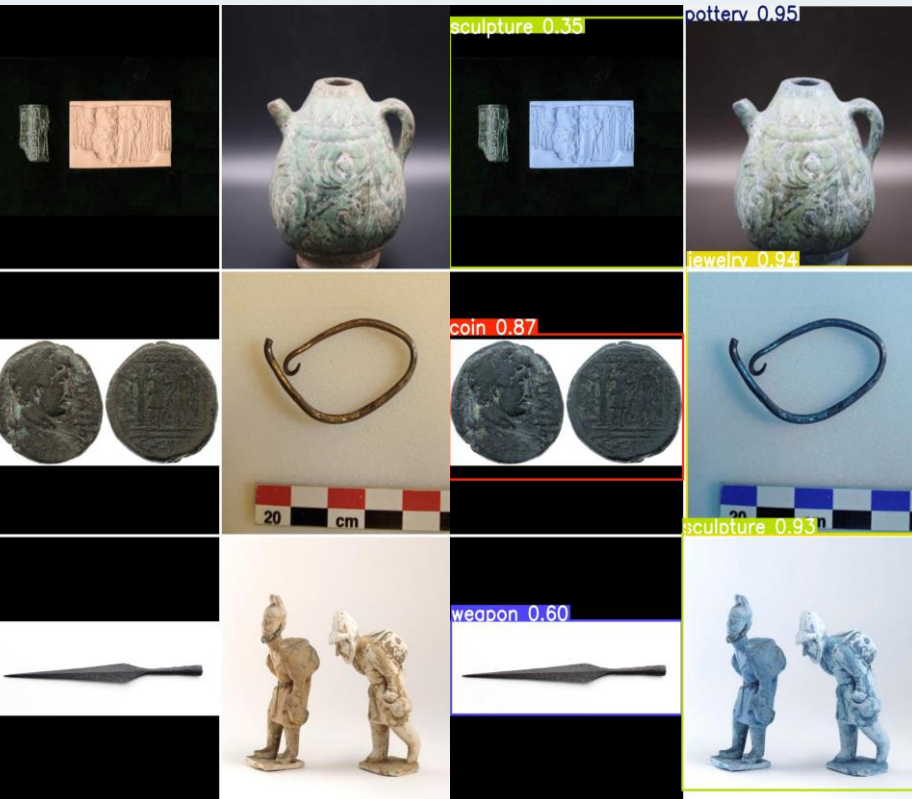
Drag and drop file here

Limit 200MB per file - JPG, JPEG, PNG



3600.png 157.9KB

Image uploaded successfully! Processing...



AI Driven Archaeological Site Mapping

Artifact Detection (YOLOv5) [Site Mapping \(U-Net\)](#) Erosion Prediction

Site Mapping Segmentation

Choose an image for segmentation



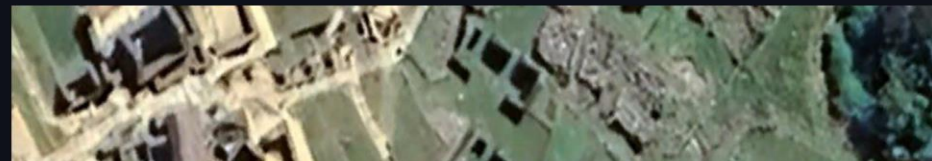
Drag and drop file here

Limit 200MB per file - JPG, JPEG, PNG

Browse files



ruin13.png 15.5MB



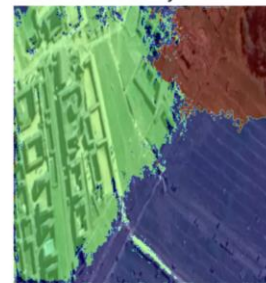
Original Image



Predicted Mask



Overlay



Class Distribution

Background
36.57%

Ruin
47.97%

Vegetation
15.46%

Presentation Video...

https://drive.google.com/file/d/1XII0v7jr2nGM-1PSZFaH-kwtJRLxhaWQ/view?usp=drive_link



Thank you

Q & A
